

Sensorless Control for Surface Mounted PM Machine with a High Inertial Load

Qidong Li, Xudong Wang, Jia Jiang, Qiao Zhang, *Member, IEEE*, Qiaoling Tong, *Member, IEEE*

Abstract—PM machine or Permanent magnet synchronous motor (PMSM) is a nonlinear system with multivariable couplings. To achieve the sensorless control of a PMSM with high inertial load, a modified current observer, using PI regulator instead of sliding mode switching function, is proposed in this paper. The modified current observer can solve the chattering and phase delay problem while still maintaining the robust advantages of sliding mode system in position estimation. In addition, a new phase-locked loop (PLL) based angle switching strategy is designed to ensure the motor can smoothly switch from I-F control to closed-loop sensorless vector control in startup stage with a high inertial load. The simulation and experimental results show that the control system of PMSM with proposed ideas has fast response speed, accurate rotor position estimation, stable state switching and good system robustness under high inertia load.

Index Terms—Back-EMF, high inertial load, PLL, PMSM, sensorless control.

I. INTRODUCTION

PMSMs have simple structure, small size, high torque inertia ratio, high energy density, high operating efficiency and adaptability to the environment and other characteristics, which makes the permanent magnet synchronous motor drive gradually replace the traditional DC drive, widely used in household electrical appliances, industrial drives, automotive traction and other fields.

Unlike DC motors, PMSMs are controlled in different ways [1-3]. There is a coupling between the flux excitation current component and the torque component of the PMSMs [4]. In order to make the PMSM control strategies have the same advantages as the DC motors, the PMSM must be decoupled. For PMSMs, the commonly used method to achieve decoupling is the use of vector transformation for magnetic field orientation control. Since there is no direct axis current, the motor does not have a direct axis armature reaction which demagnetizes the permanent magnet. All the current injected to the motor is used to generate electromagnetic torque, this can make the current control more efficient.

To get the magnetic field orientation control accurately, the position of the rotor must be precisely detected. The

conventional PMSM control system usually adopts the position sensor to obtain the position information. This leads to high cost and low reliability. In order to overcome a series of defects of the position sensors, sensorless control which is a reliable and low-cost control method has become a research hot spot in the field of motor control technology.

With the development of sensorless control technology, its applications become more and more widely. The sliding mode variable structure current observer is the most used robust method for its parameter independence characteristic. However, the chattering problem can degrade dramatically the system stability, especially in low-speed range. In order to reduce the chattering problem, higher order sliding modes are used [5]. Whereas the complexity of their implementation is high. Second Order Sliding Mode has well trade-off between chattering elimination and computing time. Super-twisting algorithm (STA) is widely employed in observers and controllers [6]. However, constant sliding-mode coefficients are adopted in their implementation, which limits the speed range of high-performance sensorless operation. In order to extend speed control range, adaptive STA-SMO is used for online tuning of the sliding mode coefficients [7-8].

Since the high inertial load increases system instability, chattering problem of sliding mode variable structure can reduce system unreliability. In this paper, a novel current observer scheme is proposed. In which the switching function of sliding mode variable structure is replaced by a PI regulator, this can not only keep the sliding mode variable structure system robustness but also eliminate system chattering to a great extent.

Another issue for current observer based sensorless control strategies is that it can work well at middle and high-speed ranges, whereas this kind of methods tend to fail at low speed and standstill operations due to unobserved machine model in that case. Therefore, specific starting up techniques should be employed for wide speed range operation. Method based on V-F control are presented in [9]. However, V-F control method makes the system sensitive to motor parameters. In order to start the motor smoothly and robustly, I-F control strategy has been employed [10]. The transient between I-F control and BEMF based close-loop control brings large disturbance in current and BEMF estimation unless a smooth transition is employed. In [10], a first-order lag compensator is employed to ensure a transition to FOC. However, the transient gain must be carefully designed. Other transition methods have been proposed

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Qidong Li is with the School of Optical and Electronic Information, Huazhong University of Science and Technology (e-mail: 760113168@qq.com).

Xudong Wang is with the School of Automation, Wuhan University of Technology (e-mail: 514771075@qq.com).

to overcome this obstacle [11-12].

In this paper, I-F control scheme is chosen for starting up operation, while a PLL composed of PI regulator and integrator is employed to acquire smooth transition between I-F control and FOC.

In this paper, Section II reviews the basic principle for PMSM control, including field oriented control and angle estimation. Section III focuses on some control methods for system implementation, and introduces new strategy for state switching. Section IV proves the improvement of the new control strategy by comparing with the conventional approach on the Simulink. Section V sets up experiments to prove the feasibility of this paper design.

II. PRINCIPLE FOR PMSM CONTROL

A. Vector Control

The PMSM vector control is based on the rotation coordinate transformation and the electromagnetic torque equation of the motor. By using Park conversion, the three-phase current can be decoupled to d, q axis current.

The specific transformation follows:

$$\begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix} = \frac{2}{3} \begin{bmatrix} \cos(-\omega t) & \cos\left(\frac{2\pi}{3}-\omega t\right) & \cos\left(\frac{4\pi}{3}-\omega t\right) \\ \sin(-\omega t) & \sin\left(\frac{2\pi}{3}-\omega t\right) & \sin\left(\frac{4\pi}{3}-\omega t\right) \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} i_A \\ i_B \\ i_C \end{bmatrix} \quad (1)$$

i_d is aligned with the direction of enhancing rotor flux linkage, while i_q is orthogonal with i_d and leading it.

The flux and voltage equations in rotating reference frame are as follows:

$$\begin{bmatrix} \varphi_d \\ \varphi_q \end{bmatrix} = \begin{bmatrix} L_d & 0 \\ 0 & L_q \end{bmatrix} \begin{bmatrix} i_d \\ i_q \end{bmatrix} + \begin{bmatrix} \varphi_f \\ 0 \end{bmatrix} \quad (2)$$

$$\begin{bmatrix} u_d \\ u_q \end{bmatrix} = R_s \begin{bmatrix} i_d \\ i_q \end{bmatrix} + \frac{d}{dt} \begin{bmatrix} \varphi_d \\ \varphi_q \end{bmatrix} + \omega \begin{bmatrix} -\varphi_q \\ \varphi_d \end{bmatrix} \quad (3)$$

Where R_s is the phase winding resistance, ω indicates the rotor angular speed, while φ_d , φ_q represent the d-axis and q-axis flux linkages respectively. L_d , L_q represent the d-axis and q-axis flux linkages respectively.

For the PMSMs, the electromagnetic torque equation is shown in equation (4) and (5).

$$\vec{T} = \frac{3}{2} P_n \begin{bmatrix} \varphi_d \\ \varphi_q \end{bmatrix} \times \begin{bmatrix} i_d \\ i_q \end{bmatrix} \quad (4)$$

$$T = \frac{3}{2} P_n [\varphi_f i_q + (L_d - L_q) i_d i_q] \quad (5)$$

Where P_n donates the pole number of motor, $\vec{T}$ and T represent the electromagnetic torque in vector form and scalar form respectively.

For surface mount motor, $L_d = L_q$, the above equation can be re-written as ,

$$T = \frac{3}{2} P_n \varphi_f i_q \quad (6)$$

Where φ_f donates the rotor magnetic flux.

As equation (6) shows, surface mount PM motor can operate the same as a DC motor by applying Park conversion. [9] Therefore, the rotor position is a critical concern for PMSM's high efficient running.

B. Rotor Position Estimation

In order to acquire rotor position without adding any position sensor, diverse sensorless control methods has been proposed which can be classified as fundamental model based method and machine saliency based method.

The fundamental model based method extracts the information for rotor position and speed through a state observer measuring fundamental excitation variables such as stator voltages and currents. However, the BEMF or magnetic flux estimated by state observer is in a positive proportion to speed. In the middle and high speed range, the BEMF or magnetic flux is large enough to be distinguished from noise. In that case, fundamental model based method can acquire the accurate position information simplicity and straightforwardness. In low speed or standstill cases, the signal-to-noise ratio of BEMF or magnetic flux is too low to extract accurate rotor angular. On the other hand, the saliency based method shows great advantages at low speed, especially at standstill. The saliency based method detects the saliency of the machine which is independent with speed. Whereas that method needs a certain saliency property of the PMSM. Therefore, for surface mounted PM motor, it is more inclined to use the fundamental based method instead of the saliency based method.

For the reason that fundamental based method meets failure at low speed, an open loop method is used to drag the motor to a certain position under proper speed. Once the motor reaching the setting speed, a novel state-switch strategy is proposed to devolve the control on back-EMF method. The main point of this paper is sensorless control by back-EMF of a surface mounted PM machine working in high speed range with high inertia load.

The back-EMF angle estimation is established on the stationary reference frame. The mathematical model is given as follow.

Through Clarke conversion, magnetic chain and voltage equation of the fixed frame is represented as

$$\begin{bmatrix} i_\alpha \\ i_\beta \\ i_0 \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} i_A \\ i_B \\ i_C \end{bmatrix} \quad (7)$$

$$\begin{bmatrix} \varphi_\alpha \\ \varphi_\beta \end{bmatrix} = \begin{bmatrix} L_\alpha & 0 \\ 0 & L_\beta \end{bmatrix} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} + \varphi_f \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix} \quad (8)$$

$$\begin{bmatrix} u_\alpha \\ u_\beta \end{bmatrix} = R \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} + \frac{d}{dt} \begin{bmatrix} L_\alpha & 0 \\ 0 & L_\beta \end{bmatrix} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} + \omega \varphi_f \begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix} \quad (9)$$

The position information of the motor is contained in the third term of the equation (9), and the position of the rotor can be obtained by solving the inverse trigonometric function. Whereas the BEMF estimated directly by that open-loop method varies from the parameters of motor, which means open-loop method is restricted in limited speed range.

The sliding mode variable structure control scheme shows great advantages in terms of parameter insensitive control with high robustness.

The control mathematical model of sliding mode variable structure is given as follows

$$\begin{bmatrix} U_\alpha \\ U_\beta \end{bmatrix} = A \begin{bmatrix} \hat{I}_\alpha \\ \hat{I}_\beta \end{bmatrix} + B \frac{d}{dt} \begin{bmatrix} \hat{I}_\alpha \\ \hat{I}_\beta \end{bmatrix} + K \begin{bmatrix} \text{sgn}(\hat{I}_\alpha - I_\alpha) \\ \text{sgn}(\hat{I}_\beta - I_\beta) \end{bmatrix} \quad (10)$$

Where I_α and I_β are the actual sampled currents, while $\hat{I}_\alpha$ and $\hat{I}_\beta$ are the estimated currents. A, B and K represent the coefficient matrix, respectively. The estimated back EMFs, E_α and E_β , are filtered from the signum function.

However, the use of a signum function will inevitably introduce high frequency ripple, which significantly deteriorates control effect. Although adding a filter may reduce the influence of chatter on the position estimation, there is a tradeoff between the filter bandwidth and the delay time. Phase compensation can be incorporated into the solution to overcome the delay time, whereas it increases system complexity. In order to eliminate the chatter, a PI controller is used to replace the signum function. The improved mathematical model is as follows:

$$\begin{bmatrix} U_\alpha \\ U_\beta \end{bmatrix} = A \begin{bmatrix} \hat{I}_\alpha \\ \hat{I}_\beta \end{bmatrix} + B \frac{d}{dt} \begin{bmatrix} \hat{I}_\alpha \\ \hat{I}_\beta \end{bmatrix} + K \begin{bmatrix} \text{PI}(\hat{I}_\alpha - I_\alpha) \\ \text{PI}(\hat{I}_\beta - I_\beta) \end{bmatrix} \quad (11)$$

Using PI regulator instead of signum function significantly reduces the chatter problem, while still maintaining the advantage of parameters insensitive. The BEMFs, ε_α and ε_β , are estimated by the output of PI regulator.

$$\begin{bmatrix} \varepsilon_\alpha \\ \varepsilon_\beta \end{bmatrix} = K \begin{bmatrix} \text{PI}(\hat{I}_\alpha - I_\alpha) \\ \text{PI}(\hat{I}_\beta - I_\beta) \end{bmatrix} \quad (12)$$

$$\begin{bmatrix} \varepsilon_\alpha \\ \varepsilon_\beta \end{bmatrix} = \omega \varphi_f \begin{bmatrix} -\sin(\theta) \\ \cos(\theta) \end{bmatrix} \quad (13)$$

The position can be obtained by inverse trigonometric operation. The specific control diagram is shown as Fig. 1

The estimated current is obtained by (14):

$$\hat{I} = \frac{1}{L} \int [U - \hat{I}R - \text{PI}(\hat{I} - I)] dt \quad (14)$$

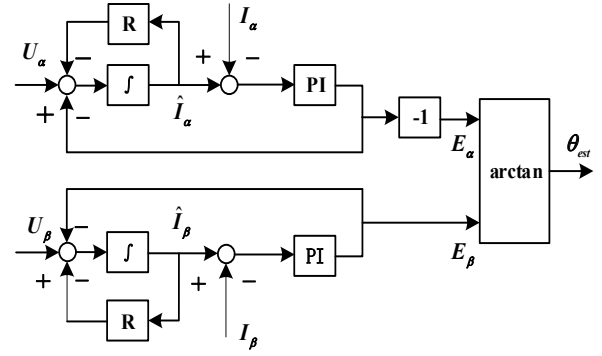


Fig. 1. The Back-EMF specific control flow chart.

The negative feedback of estimated current can eliminate the cumulative error of integration. The position extracted from the estimated BEMF is used as the reference of PLL.

The complete position estimation control diagram is given as Fig. 2.

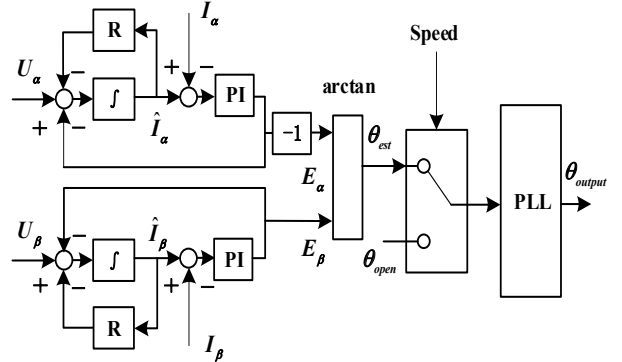


Fig. 2. The complete position estimation control diagram.

III. STRATEGY FOR SYSTEM IMPLEMENTATION

A. If Control Self-Starting Strategy

Due to the failure that BEMF met in low speed range, a start-up strategy should be applied in surface mounted PM machine control system. Conventionally, the start-up methods include VF control and IF control. Comparing with VF control, IF control method is more appropriate to the current close-loop control after BEMF has taken effect. Since the only difference between IF control and FOC is the reference of the PI regulator of i_q , a constant value for IF control and the output of speed PI regulator for FOC, respectively.

For conventional IF start-up control method, a large current is used to stall the rotor at a fixed start position. After that, the motor accelerates to a certain speed by open-loop control. However, a large start-up current is not suitable for PM machine with large inertia. Since large current results in large oscillation amplitude, and high inertia load means long stabilization time, which both makes the motor stabilize slower. In order to reduce the oscillation and accelerate the motor start-up process, the current reference value should be in a proper range.

feedback to BEMF observer, which makes motor running worse.

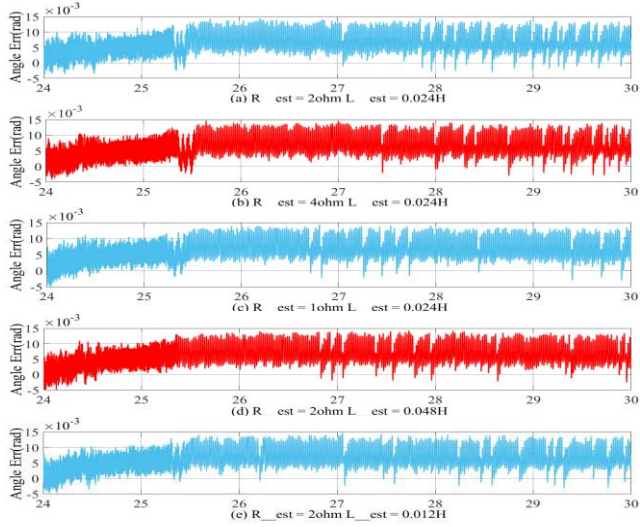


Fig. 6. Effect of parameter variation on the position estimation accuracy.

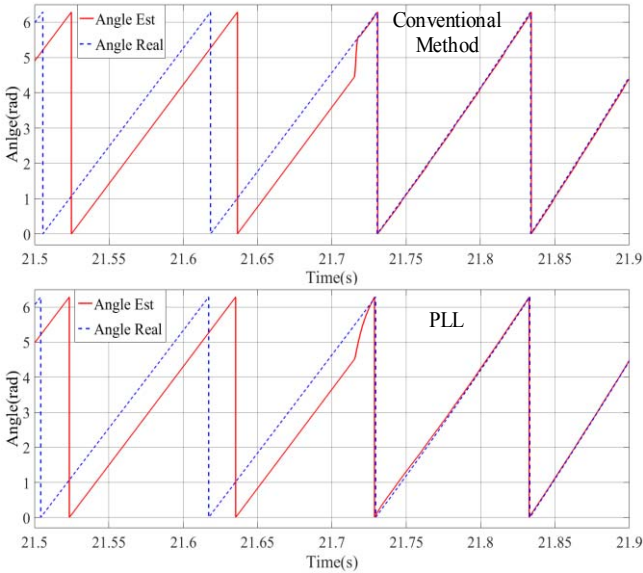


Fig. 7. The angle simulation waveform comparison

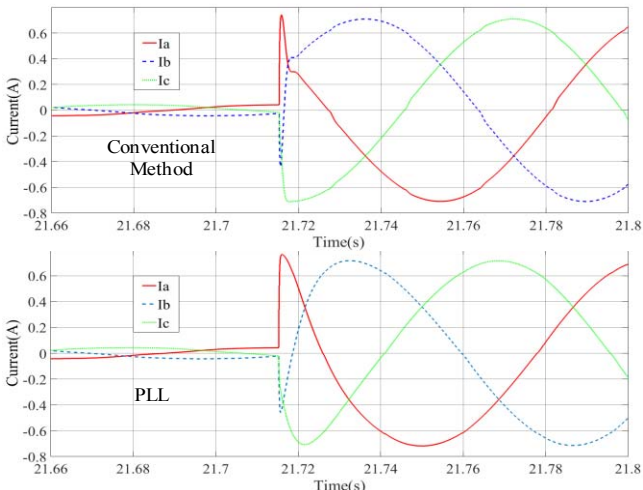


Fig. 8. The three-phase current simulation waveform comparison

As the results of Fig. 7 and Fig. 8, the transition strategy of PLL is much smoother than the conventional PI angle switching. The PM control system becomes much more robust.

Fig. 8 shows that the real speed of the motor fluctuates slightly around the theoretical open-loop speed in the process of IF accelerating. This phenomenon represents the effectiveness that high inertia load introduces. Since high inertia load results in long stabilization time.

In the closed-loop control process, the speed estimated by BEMF follows the real speed accuracy and stable in both acceleration phase and the final stabilization phase.

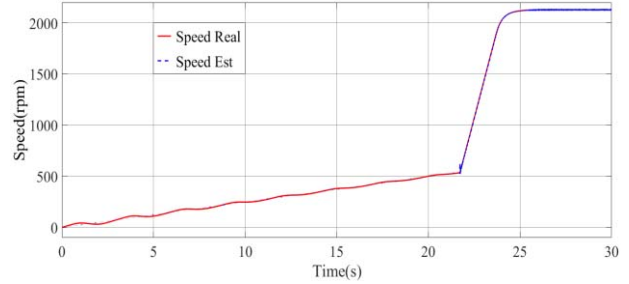


Fig. 9. Open loop motor speed.

Combined with the above waveforms and analysis, it can be seen that the rotor position and speed can be estimated accurately and feasibly by back EMF. In order to get more convincing verification of the overall design proposed in this paper, the experimental analysis is carried out in next section.

V. EXPERIMENTAL ANALYSIS

Experimental motor is a surface mounted PM machine, the motor is mounted on a large inertia fan and drives it directly. The machine pole pairs number are 5, its rated power is 300W with its rated voltage is 110V AC. Its working speed range is from 600 rpm to 1300 rpm. The experimental setting is shown in Fig.10.

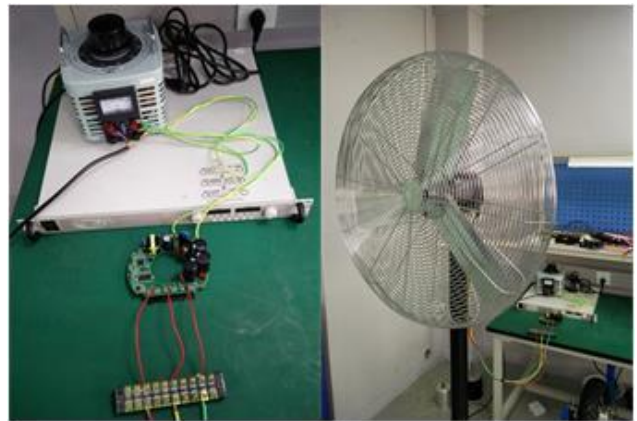


Fig.10. Experimental device.

The threshold of speed that transiting from IF control to current observer control is set to 108 r/min. The figures below show the three-phase current wave while the experimental PM machine operating under the speed of 720 rpm, 960 rpm and 1200 rpm, respectively. There is slightly distortion in three phase current waveform, for the reason of sample noise and

other nonlinear elements in the system. However, the current frequency and amplitude are generally stable. Meanwhile, the motor stays stable during the transition between IF control and BEMF control.

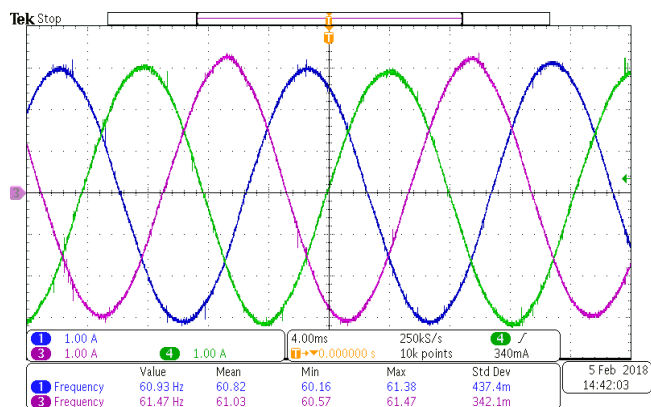


Fig. 11. 720 rpm Three-phase current waveforms

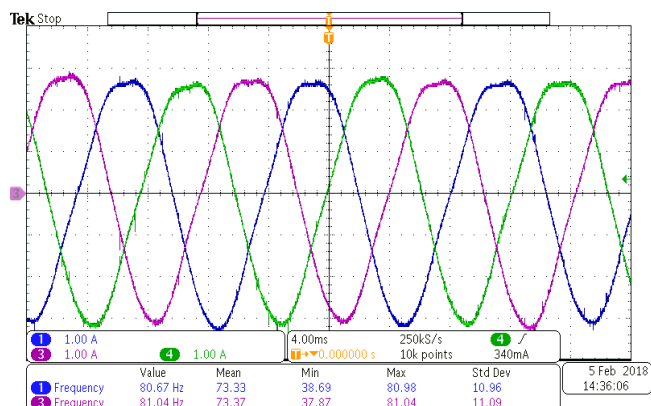


Fig. 12. 960 rpm Three-phase current waveforms.

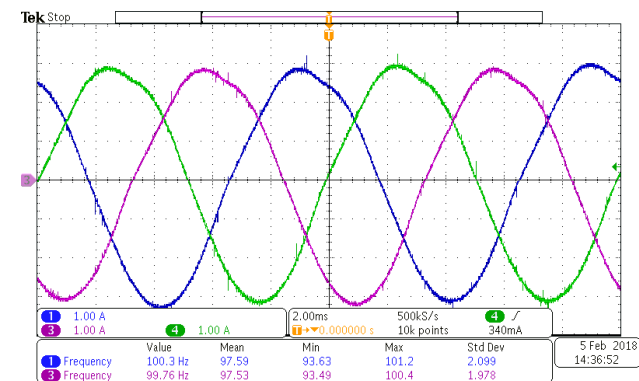


Fig. 13. 1200rpm Three-phase current waveforms.

VI. CONCLUSION

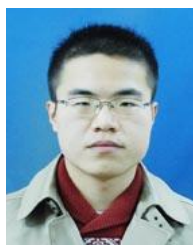
In this paper, a modified current observer in rotor position estimation has been proposed. After simulation and experimental research, draw the following conclusions:

- 1) Improved I-F self-starting strategy allows the motor to start smoothly and quickly with high inertia load.
- 2) The proposed phase-locked loop switching strategy makes motor switching from the I-F control mode to the vector control mode smoothly.

- 3) Using the PI regulator instead of the sliding mode switching function in position estimation can not only maintain the robustness of the conventional sliding mode observer but also solve the chattering problem.

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Qidong Li received a bachelor degree in electronic engineering from Huazhong University of Science and Technology, Wuhan, Hubei in 2016. He is studying for a master degree in electronic engineering at Huazhong University of Science and Technology.

His research interest is permanent magnet synchronous motor system.



Xudong Wang received a bachelor's degree in automation from Jiangnan University, Wuhan, Hubei in 2017. At present, he is studying for a master degree in electric engineering at Wuhan University of Technology.

Under the guidance of his tutor, Qiao Zhang, he carried out the research of the permanent magnet synchronous motor from 2017 to 2018.

His research interest covers reactive power compensation in automated electrical systems and permanent magnet synchronous motor systems.



Qiaoling Tong (M'12) received the B.Eng. and Ph.D. degree from the School of Optical and Electronic Information, Huazhong University of Science and Technology, China, in 2003 and 2010, respectively. From 2008 to 2010, He was a research scholar with the Department of Electrical Engineering and Computer Science, University of California, Irvine.

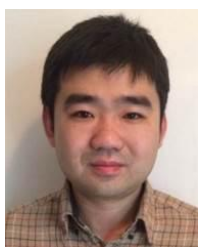
He is currently an associate professor the School of Optical and Electronic Information, Huazhong University of Science and Technology, China. His current research interests include sensorless control of DC-DC converters and VLSI



Jia Jiang is studying integrated circuit design and integration system for a bachelor degree at Huazhong University of Science and Technology Wuhan, Hubei.

In June of this year, she will study for a master degree in electronic engineering at Huazhong University of Science and Technology. Her research interest is permanent magnet synchronous motor

system.



Qiao Zhang (M'11) received the B.Eng. M.Eng and Ph.D. degrees from the Huazhong University of Science and Technology, China, in 2003, 2006 and 2010, respectively. From 2008 to 2009, He was a visiting scholar with the Department of Electronic and Electrical Engineering, University of Sheffield, UK. After that he

works as a senior research engineer in IMRA Europe S.A.S. for 7 years. He currently works in School of Automation, Wuhan University of Technology. His research interest focuses on power electronics system design and control, such as DC-DC converter sensorless control strategies, electrical machine parameters estimation by control theory, system nonlinearity compensation for DC-DC converters and voltage source inverters.